

Name: M. Samanvi

Reg. No.: 192425318

Course Code: MLA0401

Course Name: Deep Learning

1. CNN vs Traditional Neural Networks

Architecture: CNN vs ANN

CNN — Convolutional Neural Network

A CNN is a neural network specially designed for images and visual data.

It uses:

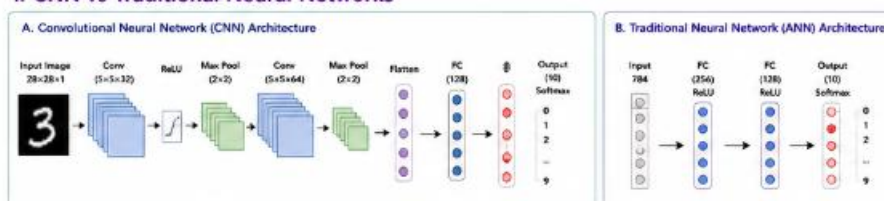
- Convolution layers → extract features like edges, shapes, textures.
- Pooling layers → reduce the size of feature maps.
- Fully connected layers → perform final classification.
- Parameter sharing → the same filter is used across different parts of the image.

Traditional Neural Network — ANN

A traditional Artificial Neural Network (ANN) mainly uses fully connected layers.

For an image, the pixels are usually converted into a 1D vector before entering the network.

1. CNN vs Traditional Neural Networks



Comparison Table

Aspect	CNN	Traditional Neural Network (ANN)
Architecture	Convolution + Pooling + Fully Connected	Mainly Fully Connected layers
Input	Can directly process image/grid data	Usually needs flattened input
Feature Extraction	Automatic	Usually requires manual feature engineering
Parameter Sharing	Yes, convolution filters are shared	No
Spatial Information	Preserves spatial relationships	Mostly loses spatial structure after flattening
Number of Parameters	Usually fewer for images	Usually very large for images
Computational Complexity	Lower for image tasks due to local connections	Higher for large images
Feature Learning	Learns hierarchical features automatically	Learns features from flattened data
Image Processing	Excellent	Not very suitable
Translation Handling	Better at recognizing features in different locations	Poorer
Best Used For	Images, videos, computer vision	Tabular data, simple classification
Example Applications	Face recognition, medical imaging, object detection	Customer prediction, numerical classification

2. AlexNet vs ResNet

Architecture: AlexNet vs ResNet

AlexNet

AlexNet is a deep CNN developed in 2012. It became famous because it achieved excellent results in image classification and helped popularize deep learning for computer vision.

- It uses convolutional layers to extract image features.
- It uses ReLU activation, which makes learning faster.
- It uses pooling to reduce the size of feature maps.
- It uses fully connected layers for final classification.

ResNet

ResNet stands for Residual Network and was introduced in 2015.

Its main innovation is the residual/skip connection.

A skip connection allows information to pass directly from an earlier layer to a later layer. This helps solve the vanishing gradient problem, which can occur when a neural network becomes very deep.

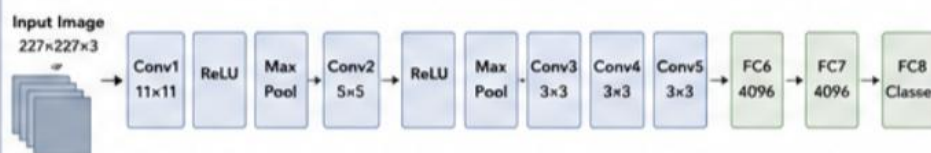
Because of this idea, ResNet can have 18, 34, 50, 101, or 152 layers and still be trained effectively.

- It uses convolutional layers for feature extraction.
- It uses ReLU activation.
- It uses residual blocks.
- Skip connections make training deep networks easier.

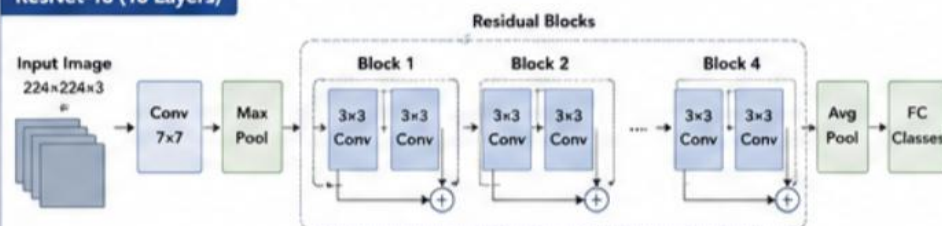
2. AlexNet vs ResNet

Architecture Diagrams

AlexNet (8 Layers)



ResNet-18 (18 Layers)



Comparison Table

Aspect	AlexNet	ResNet
Year	2012	2015
Number of Layers	8	18, 34, 50, 101, 152
Main Idea	Deep CNN for image classification	Residual learning
Skip Connections	No	Yes
Architecture	Sequential	Residual blocks
Activation	ReLU	ReLU
Training	Relatively easier	Designed for very deep training
Vanishing Gradient	Can become a problem in deeper networks	Skip connections help overcome it
Parameters	About 60 million	ResNet-50 has about 25.6 million
Computational Requirement	Lower	Higher for deeper versions
Performance	Excellent for its time	Generally higher on modern vision tasks
Depth	Shallow compared with ResNet	Very deep
Overfitting	Uses dropout	Uses architectural depth and regularization techniques
Best Use	Basic image classification	Complex image classification and computer vision
Major Advantage	Pioneer of modern deep CNNs	Allows very deep networks to train successfully
Major Limitation	Cannot easily scale to very deep networks	Deeper models need more computation

3. Max Pooling vs Average Pooling

Architecture/feature-map example

Max Pooling

Max Pooling selects the largest value from a small region of the feature map.

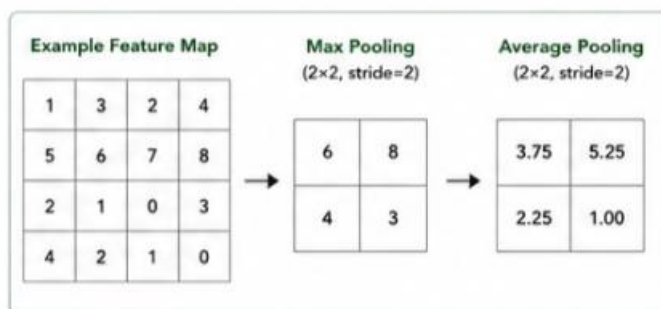
- Keeps the strongest and most important feature.
- Reduces the size of the feature map.
- Reduces computation.
- Helps detect important features such as edges and shapes.
- Less affected by small changes in the image.

Average Pooling

Average Pooling calculates the average value of all values in a small region.

- Keeps the overall information of the region.
- Produces smoother feature maps.
- Reduces the size of the feature map.
- Useful when the overall presence of a feature is important.
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3. Max Pooling vs Average Pooling



Aspect	Max Pooling	Average Pooling
Working	Selects the maximum value	Calculates the average value
Main Purpose	Keeps strongest features	Keeps overall information
Feature Selection	Selective	More distributed
Output	Strongest activation	Average activation

Aspect	Max Pooling	Average Pooling
Noise Handling	Generally more robust to small background variations	Can be affected by irrelevant values
Feature Map	More prominent features	Smoother features
Information Retained	Important/high-level features	Overall regional information
Computational Cost	Low	Low
Best For	Object detection and image classification	Smooth representations and some classification tasks
Advantage	Highlights important features	Preserves more overall context
Disadvantage	May lose some useful information	May weaken strong important features
Common Usage	Very common in CNNs	Less common than max pooling

4. CNN Regularization Techniques

Architecture: Dropout vs Batch Normalization vs Data Augmentation

Regularization is used to prevent a CNN from overfitting.

Overfitting happens when the model performs very well on training data but performs poorly on new or testing data.

Three important techniques are Dropout, Batch Normalization, and Data Augmentation.

1. Dropout

Dropout randomly turns off some neurons during training.

2. Batch Normalization

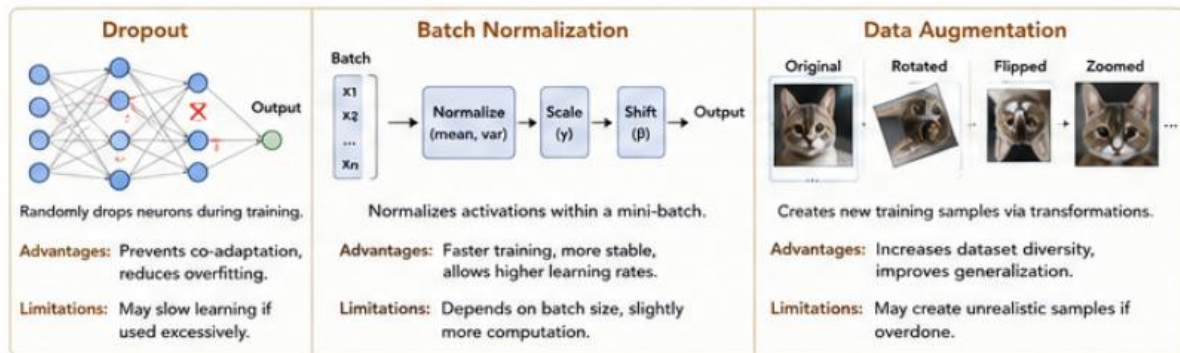
Batch Normalization normalizes the values going through a neural network layer.

It helps keep the values in a suitable range during training.

3. Data Augmentation

Data Augmentation creates new variations of existing training images.

4. CNN Regularization Techniques



Comparison Table

Aspect	Dropout	Batch Normalization	Data Augmentation
What it does	Randomly deactivates neurons	Normalizes layer activations	Creates modified training samples
Main purpose	Reduce overfitting	Stabilize and improve training	Increase data variation
Works on	Neurons	Layer activations	Input images/data
During Training	Random neurons are dropped	Normalization is applied	Modified data is provided
During Testing	Dropout is turned off	Uses learned statistics	Usually not applied
Main Benefit	Prevents neuron dependency	Stable and faster learning	Better generalization
Example	Drop 50% of neurons	Normalize mini-batch values	Rotate an image
Effect on Data	No change to input data	No change to dataset size	Increases training variations
Best Use	Large/deep networks	Deep CNN training	Small image datasets
Main Advantage	Simple and effective	Stable learning	More diverse training data
Limitation	Too much dropout can hurt learning	Adds some computational overhead	Too much augmentation can create unrealistic images

5. CNN Applications Analysis

Architecture/workflow for Classification, Detection, Face Recognition, Medical Imaging, and Autonomous Vehicles

CNNs are mainly used for image and visual data processing. They can automatically learn useful features from images and use them for classification, detection, recognition, and other tasks.

1. Image Classification

CNNs can identify what is present in an image.

2. Object Detection

CNNs can identify what objects are present and where they are located in an image.

3. Face Recognition

CNNs can learn important facial features such as:

4. Medical Image Analysis

CNNs are widely useful for analyzing medical images such as:

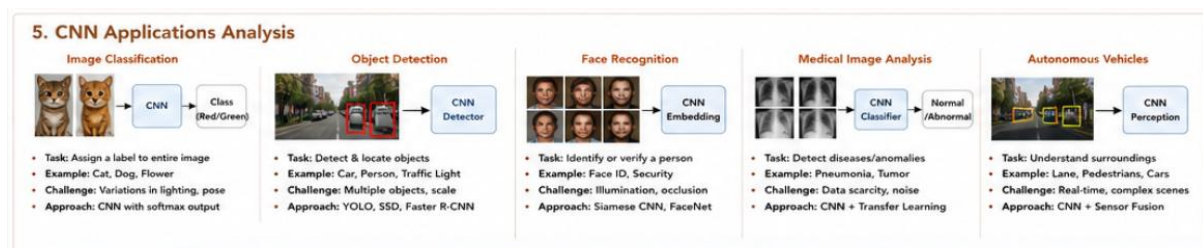
- X-rays
- MRI scans
- CT scans

5. Autonomous Vehicles

CNNs help autonomous vehicles understand their surroundings.

They can detect:

- Cars
- Pedestrians
- Traffic lights



Comparison Table of CNN Applications

Application	What CNN Does	Example
Image Classification	Identifies the class of an image	Cat / Dog
Object Detection	Identifies objects and their locations	Detect cars on a road
Face Recognition	Identifies or verifies faces	Face unlock
Medical Imaging	Detects patterns or abnormalities	X-ray analysis
Autonomous Vehicles	Understands road surroundings	Detect pedestrian and traffic sign